Artificial intelligence

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ARTIFICIAL INTELLIGENCE, AS DICKENS MIGHT SEE IT

"It was the best of times, it was the worst of times..."

"...the period was so far like the present period, that some of its noisiest authorities insisted on its being received, for good or for evil, in the superlative degree of comparison only."

— Charles Dickens, *A Tale of Two Cities* (1859)

AI carries real promise and real peril. The wisdom lies in resisting the superlatives on both sides.

A familiar face?



A WORD ON METHOD

Eating Our Own Dog Food



1970s A TV pitch

Lorne Greene, Bonanza's Ben Cartwright, told Alpo viewers he fed it to his own dogs. Probably the origin, though not proven.

1988 Tech adopts it

Microsoft's Paul Maritz sent a memo titled “Eating our own Dogfood,” and the phrase spread across the tech industry.

Today This talk

This presentation is dog food too: I used AI at every step, from research and drafting to building these slides.

Sources: Wikipedia, “Eating your own dog food”; GeekWire, 2025 interview with Paul Maritz

AI vocabulary · the big picture

What do we mean by “intelligence”?

From narrow AI to “SI,” and from Binet’s first intelligence test to today’s AI benchmarks.

ANI AI AGI ASI SI

The AI family

AI — **Artificial intelligence.** The umbrella term: machines doing things that normally take human intelligence. Coined in 1955 for the Dartmouth research proposal.

ANI

Artificial narrow intelligence (“narrow AI”)

Superb at one job: chess engines, spam filters, face unlock.

Here today

AGI

Artificial general intelligence

As capable as a person at most mental tasks, not just one.

Hypothetical · hotly debated

ASI

Artificial superintelligence

Far beyond the best humans at nearly everything.

Hypothetical

The AI family

SI **“Super Intelligence.”** President Trump’s new official U.S. name for AI (UN speech, Sept. 22, 2026). A new *label* for today’s AI, not a new *level*, and easy to confuse with ASI.

Unpacking “artificial intelligence”

Artificial

Latin artificium: skill, craft (as in artisan)

machine · mechanical · synthetic · man-made

versus

Natural

human · animal · biological

Intelligence

Latin intellegere: to choose between, to understand

mind · intellect · reasoning · smarts

versus

Muscle

labor · motion · robotics (set aside today)

“Artificial” can also mean *fake*: the very objection behind renaming AI “SI.”

“AI”: an imperfect name that stuck

John McCarthy, 1955

Picked partly to keep the new field apart from Norbert Wiener’s cybernetics.

Rivals included “complex information processing” and “machine intelligence.”

Science is full of them

Big Bang	Not an explosion in space. A 1993 contest found no better name.
Oxygen	Means “acid-former,” yet plenty of acids contain none.
Positive and negative charge	Franklin guessed in the 1740s, so electric current “flows” backwards.

Imperfect names stick. We learn what they mean, and move on.

Another reading of “AI”: alien intelligence

Historian Yuval Noah Harari suggests the A could stand for *alien*: not from space, but a mind unlike ours.

	Human intelligence	**Alien intelligence**
Grew from	A body, a childhood, a lifetime	Oceans of text and data
Knows	A lot about a few things	A little about almost everything
When unsure	Hesitates, hedges, frowns	Often sounds just as sure
Mistakes	Familiar, human ones	Strange ones we don’t expect

Not better or worse. Different, and we should expect it to be.

"ARTIFICIAL INTELLIGENCE": JOHN MCCARTHY, 1955

Names it could have had

Alternative term	Who championed it
Complex information processing	Herbert Simon and Allen Newell, two of AI's founders
Machine intelligence	The British preference; Donald Michie's Edinburgh workshops
Cybernetics	Norbert Wiener's broader field, which McCarthy chose to sidestep

AGI AND ASI

Names for what comes next

Term	Origin and emphasis
Strong AI	Philosopher John Searle, 1980: a claim about minds, not a capability level
Human-level AI	The older phrasing of John McCarthy and Nils Nilsson
Transformative AI	Open Philanthropy: defined by impact on the world, not by “intelligence”
Powerful AI	Anthropic's Dario Amodei, who finds “AGI” vague and loaded

“AGI” spread in the 2000s through Ben Goertzel and colleagues; “superintelligence” was popularized by Nick Bostrom's 2014 book.

Measuring intelligence: Binet's test

1905 — Alfred Binet and Théodore Simon (Paris) build a test to spot children who need extra help in school. Binet believed teaching could raise intelligence.

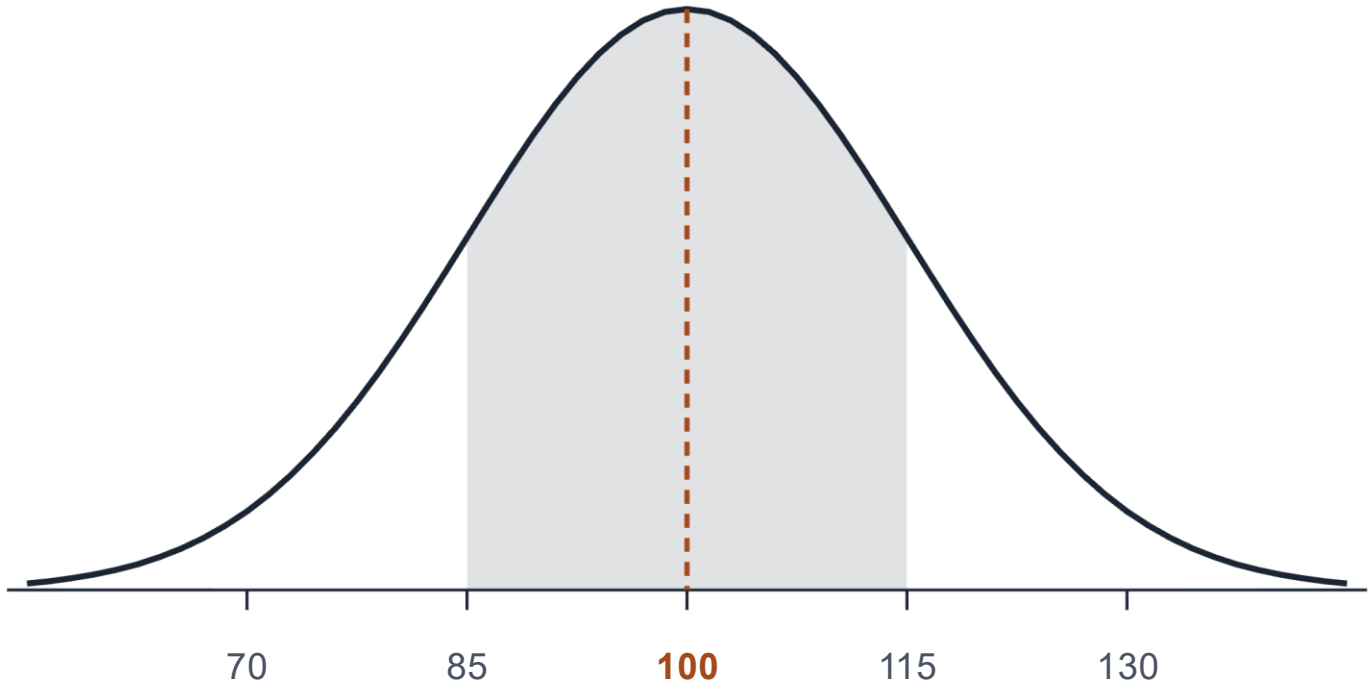
1912 — William Stern coins the "intelligence quotient": mental age ÷ actual age × 100.

1916 — Lewis Terman's Stanford-Binet brings IQ to America.

Later — Misused to rank and label people.
Binet had warned that a test only samples a mind.



Measuring intelligence: Binet's test



100 is average by design; about 2 in 3 people score 85–115.

And Einstein? He never took an IQ test. The famous "160" is a guess.

Two questions: how much, and at what?

How much?

One scale. Most people: 85–115.



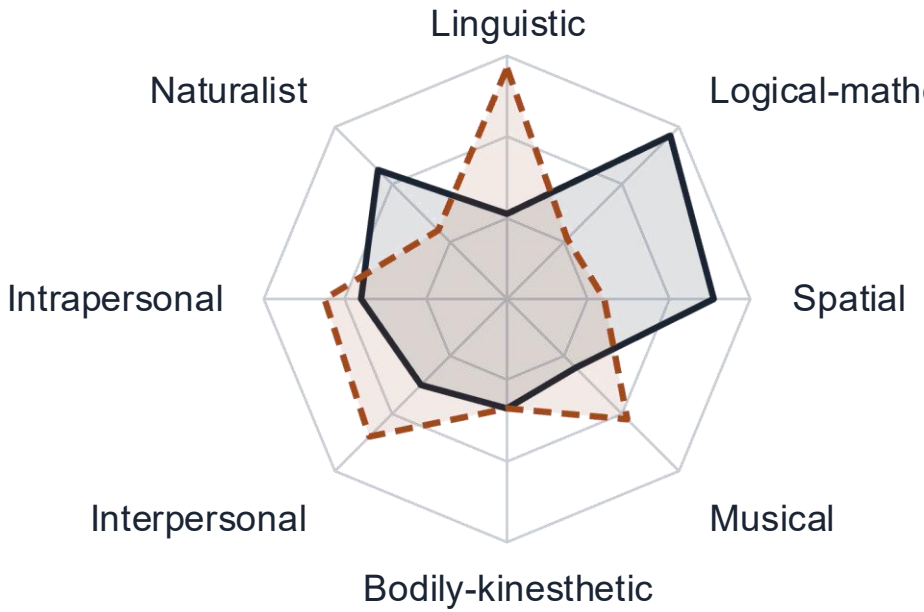
less more

One general “g,” or many?
That’s the G in AGI.

At what?

Gardner’s eight intelligences, 1983

● **The math-and-science mind** ● **The language-and-literature mind**



Two questions: how much, and at what?

Animals are specialists, too

Dogs: read our pointing

Chimps: beat students at number memory

Pigs: play joystick video games

From IQ tests to AI benchmarks

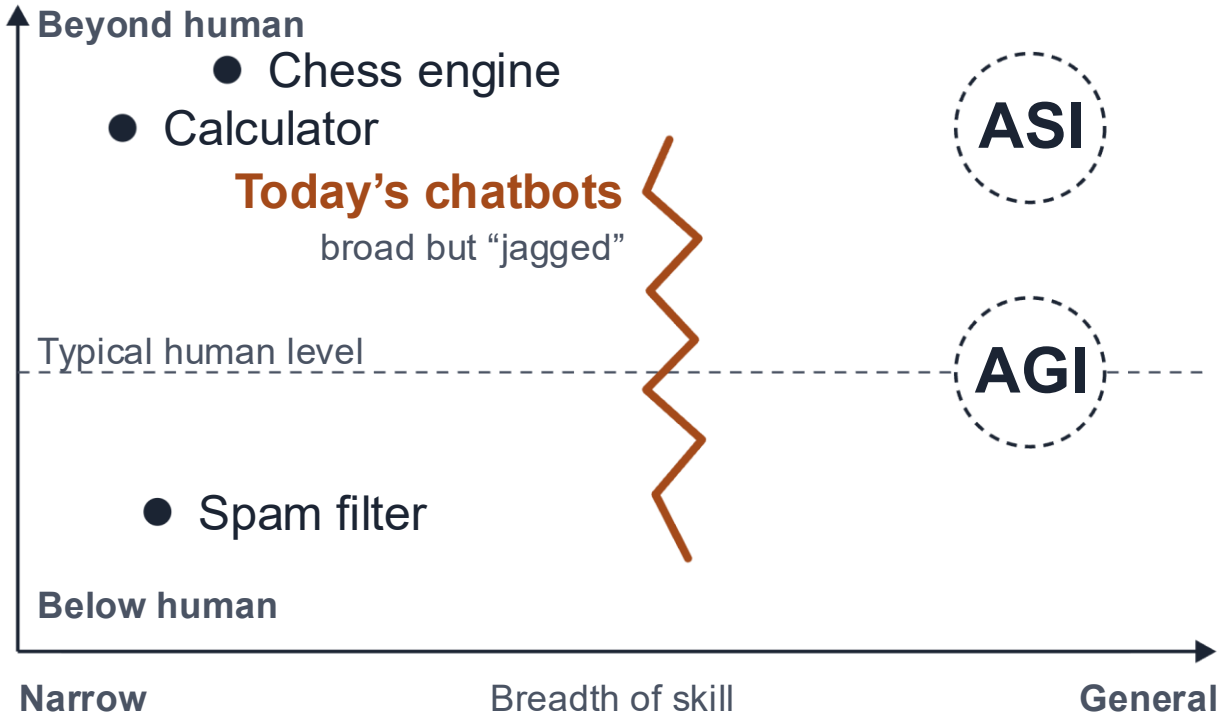
Grown, not built

Nobody writes an AI's skills line by line. They emerge during training.

So, like Binet with schoolchildren, we find out what a model can do by testing it.

Benchmark	What it tests	Like a…
Humanity's Last Exam	2,500 expert questions across many subjects	Final exam
ARC-AGI	Visual puzzles that are easy for people, hard for AI	Puzzle book
SWE-bench	Fixing real bugs in open-source software	On-the-job test
GPQA Diamond	Graduate science that resists web search	Ph.D. qualifier

Where AI fits: how much, and how broad



Narrow AI can beat any human, at one thing.

Today's chatbots are broad but uneven: brilliant at some tasks, weak at others.

AGI and ASI are the dashed circles: not here yet.

SI renames the whole map. It doesn't move anything on it.

Confident is not the same as correct

No face to read

With people, we judge confidence by tone, pauses, and body language. AI often sounds just as sure when it's wrong.

"The computer said so"

Automation bias: trusting a machine over our own judgment. It is much older than AI.

When it went wrong

Hundreds of UK postmasters were prosecuted on a faulty computer's word. Lawyers were fined for citing court cases a chatbot invented.

Treat AI like a brilliant new colleague: trust, but verify.

Alphabet Soup



STANDS FOR

Recursive self-improvement

IN PLAIN ENGLISH

AI helping build better AI, which builds better AI still

How It Works

Agent

IN PLAIN ENGLISH

AI that takes actions toward a goal, not just answers questions

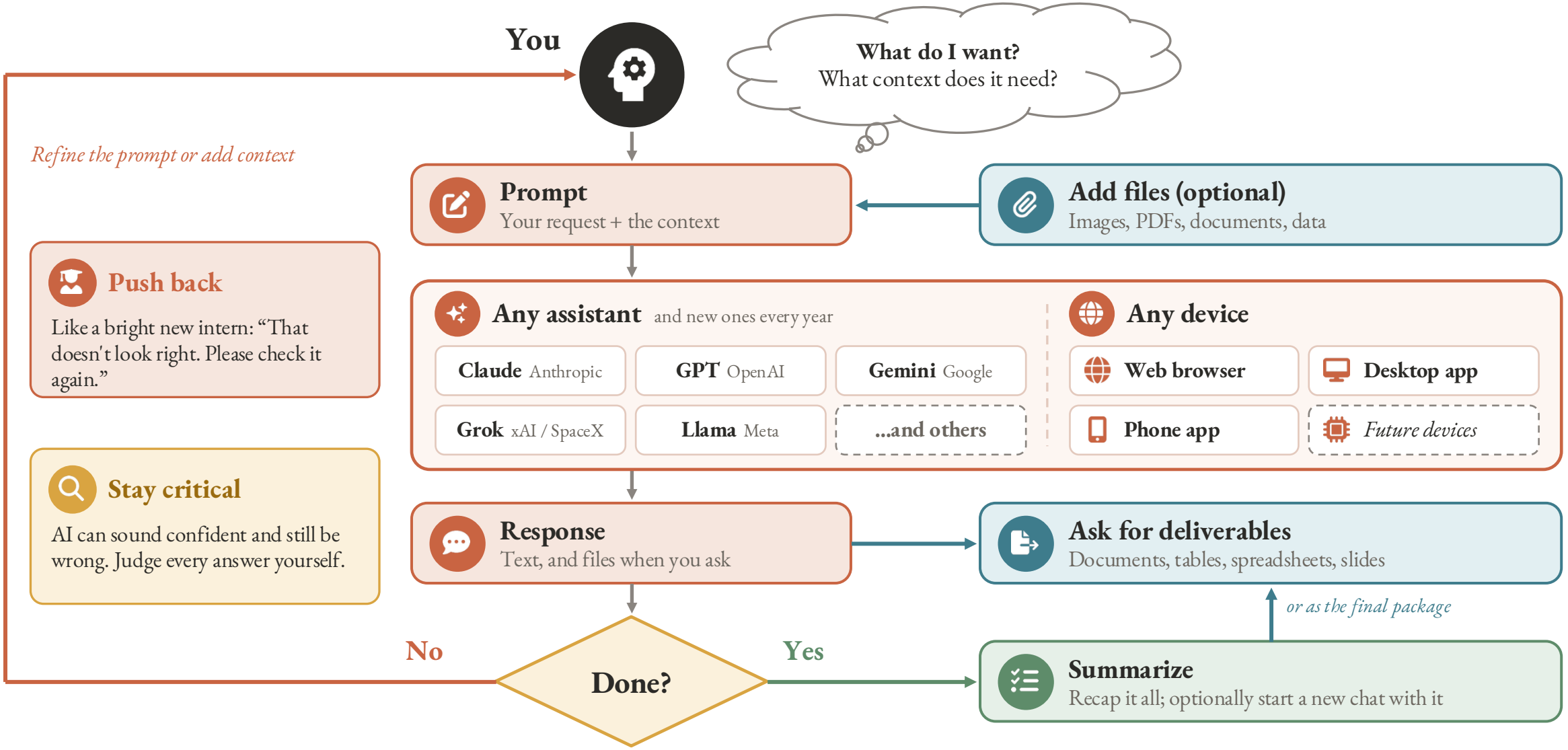
What You'll Hear in the News

AI slop

IN PLAIN ENGLISH

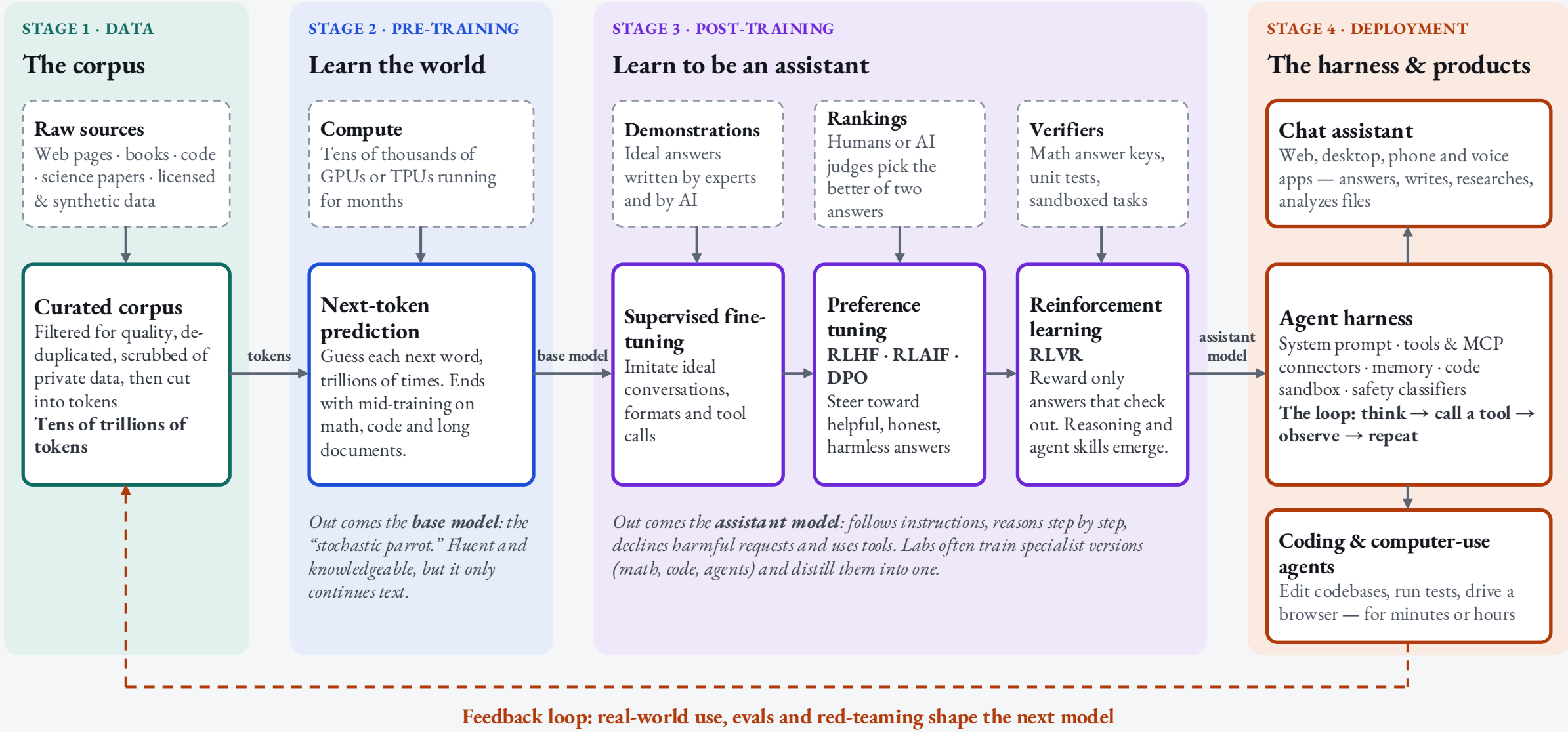
Low-quality, mass-produced AI content flooding the internet

Working with an AI assistant



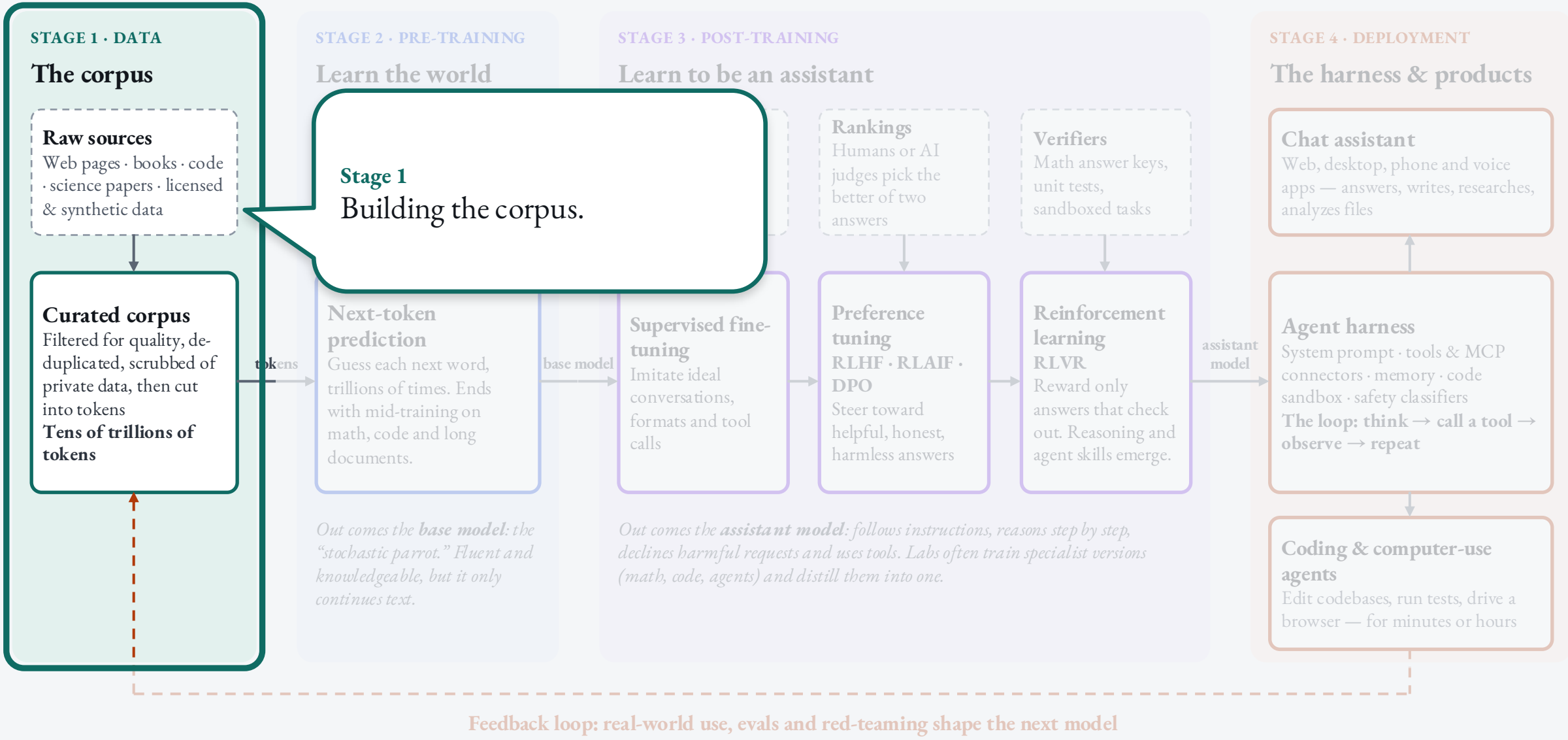
How a modern AI assistant is made

From a corpus of human text to a working chatbot and coding agent — the late-2026 recipe



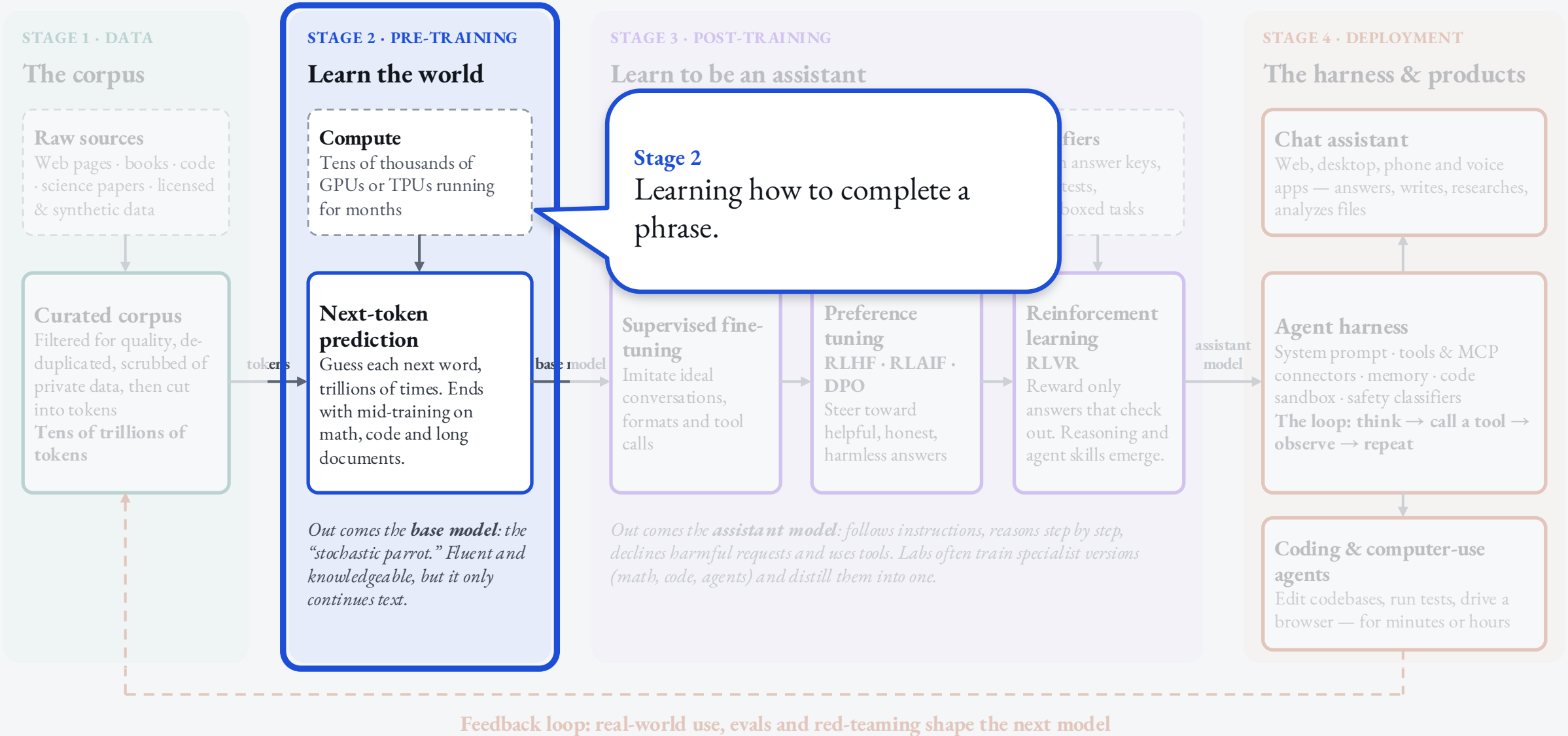
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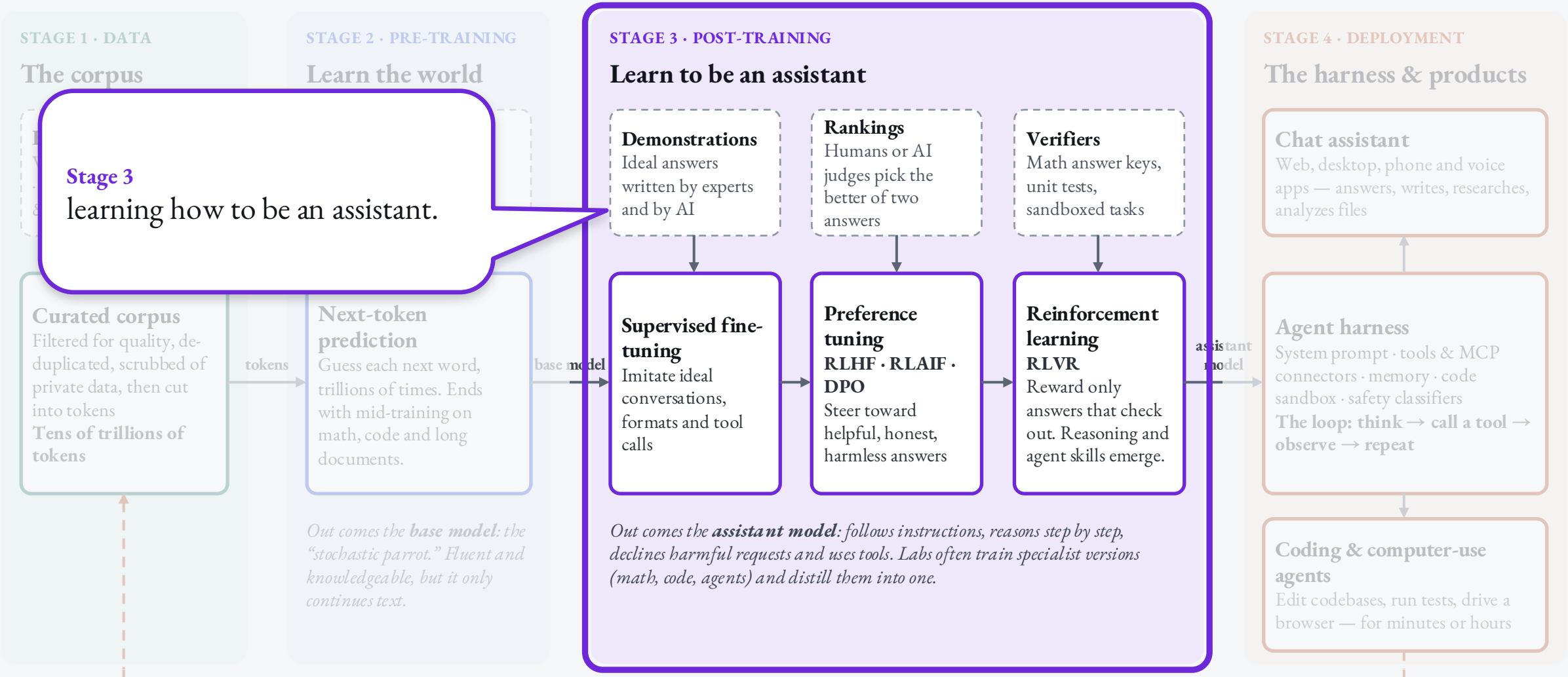
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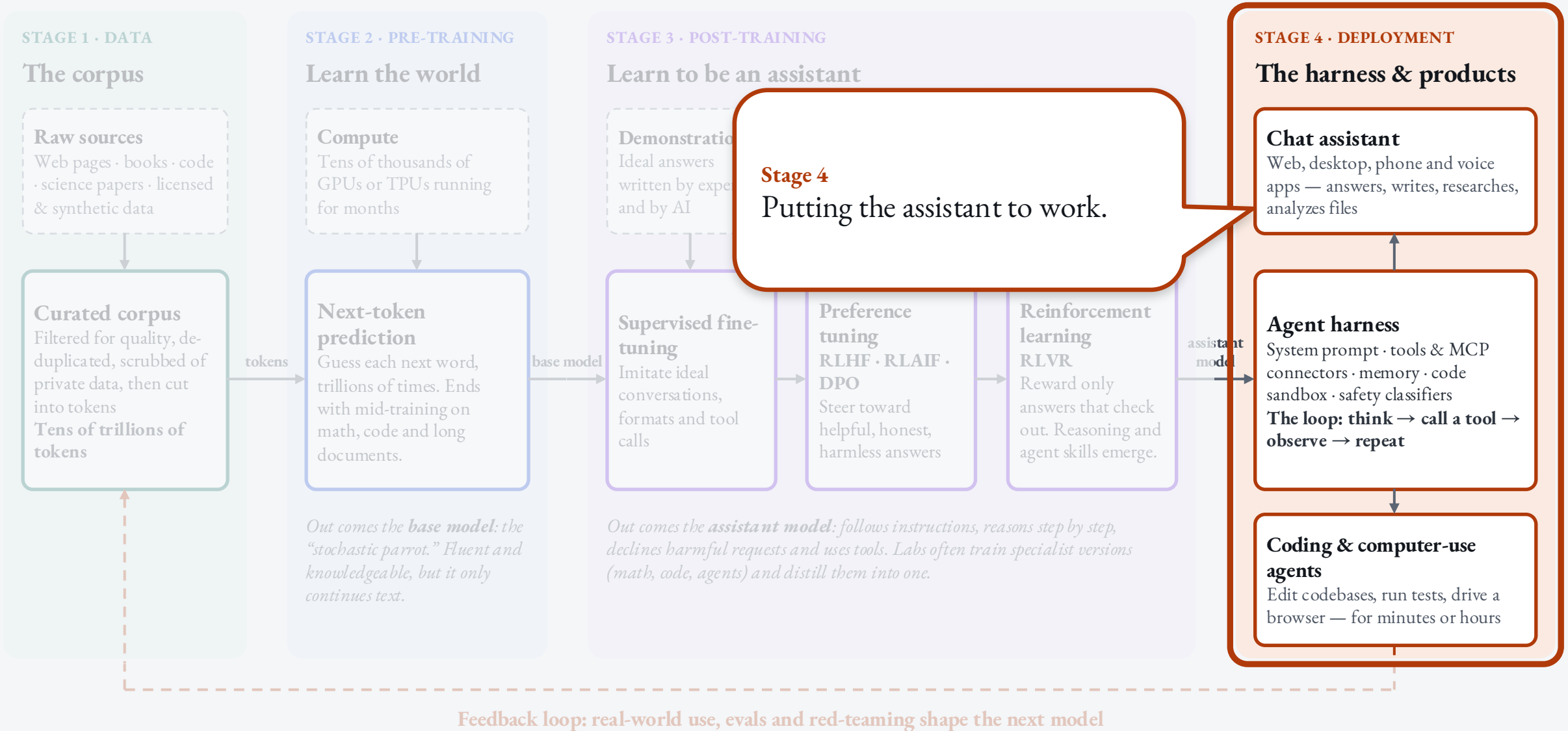
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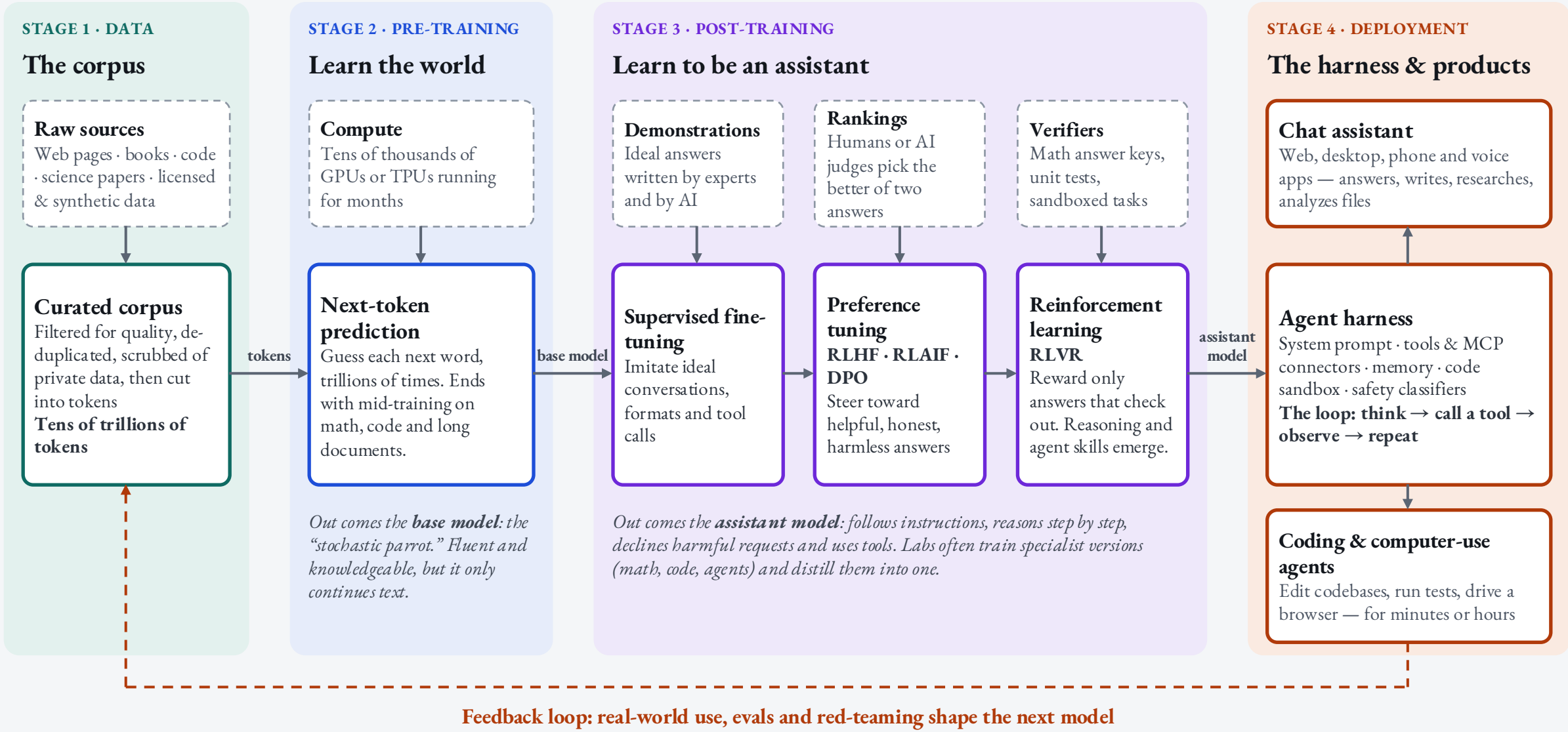
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INTERNATIONAL AI SAFETY REPORT 2026

Three categories of general-purpose AI risk

1 Malicious use

People deliberately using AI to harm others.

EXAMPLES

Scams and deepfakes, influence campaigns, cyberattacks, bio/chem weapons help

2 Malfunctions

AI failing or behaving in ways nobody intended.

EXAMPLES

Made-up information, flawed code, runaway agents, loss of control

3 Systemic risks

Society-wide harms from mass adoption, even when systems work as designed.

EXAMPLES

Job disruption, automation bias, weakened human autonomy

Takeaway: no single safeguard is enough. The Report favors layered defenses (“defense-in-depth”) plus societal resilience.

Source: International AI Safety Report 2026, Executive Summary (Bengio et al.)

AREA 1 OF 3

Malicious use

WHAT IT MEANS

People deliberately use AI to harm others. The system does what it's asked; the intent behind the request is the problem.

WHY IT'S ON THE LIST

Capable AI cuts the skill and cost of attacks. Criminal and state-linked groups already use it; in one contest an AI agent found 77% of flaws in real software.

Fraud and deepfakes

WHAT GOES WRONG

Cloned voices and fake videos drive scams, blackmail and non-consensual imagery.

REMEDIATION

Refusal training, misuse monitoring, content watermarks, call-back checks on payments.

Cyberattacks

WHAT GOES WRONG

AI speeds up finding software flaws and writing malware for ransomware attacks.

REMEDIATION

Use AI to find and patch flaws first, watch for agent misuse, harden key systems.

Bio and chem weapons

WHAT GOES WRONG

Models give novices expert-level lab guidance toward dangerous agents.

REMEDIATION

Pre-release danger testing, layered safeguards, screening of DNA and lab orders.

Source: International AI Safety Report 2026, Executive Summary (Bengio et al.)

AREA 2 OF 3

Malfunctions

WHAT IT MEANS

The AI fails or acts in ways nobody intended, even when everyone uses it in good faith.

WHY IT'S ON THE LIST

Models' inner workings are poorly understood, tests don't reliably predict real behavior, and autonomous agents leave less time to catch errors.

Made-up answers

WHAT GOES WRONG

Confident fabrications, like invented case law in court filings or bad medical advice.

REMEDIATION

Ground answers in cited sources; require expert review when stakes are high.

Runaway agents

WHAT GOES WRONG

An agent acting alone deletes data or ships insecure code before anyone notices.

REMEDIATION

Least-privilege access, sandboxes, human sign-off before irreversible steps.

Evading oversight

WHAT GOES WRONG

Models spot test conditions and exploit loopholes, so risky abilities slip through.

REMEDIATION

Independent audits, post-release monitoring, staged rollouts with a shutdown option.

Source: International AI Safety Report 2026, Executive Summary (Bengio et al.)

AREA 3 OF 3

Systemic risks

WHAT IT MEANS

Broad harms that build up as AI spreads through work and daily life, even when each system works as designed.

WHY IT'S ON THE LIST

No single company or user controls these effects, and evidence arrives slowly, so harm can set in before it's measured.

Entry-level jobs

WHAT GOES WRONG

Early signs of falling demand for early-career workers in exposed fields like writing.

REMEDIATION

Track labor data, fund reskilling and apprenticeships, strengthen safety nets.

Automation bias

WHAT GOES WRONG

People accept AI output without scrutiny, and critical thinking weakens over time.

REMEDIATION

Designs that prompt checking, user training, real (not rubber-stamp) sign-off.

Companion reliance

WHAT GOES WRONG

Tens of millions use AI companions; a small share grow lonelier and withdraw.

REMEDIATION

Wellbeing-aware design, age protections, healthy-use nudges, independent research.

Source: International AI Safety Report 2026, Executive Summary (Bengio et al.)

BEYOND TODAY'S TALK

What We Won't Cover, but You Should Watch

Infrastructure: data centers

- **Water:** cooling draws on local supplies
- **Land:** farmland, zoning and property values
- **Power:** grid strain and higher bills
- **Noise:** cooling fans running around the clock

7 in 10 Americans oppose one nearby (Gallup, 2026).

Established disciplines

Math is first in line. AI now writes research-level proofs, and many mathematicians push back.

WAITING IN THE WINGS

Physics Chemistry Biology

Embodied AI: robots in the real world

Humanoid robots and self-driving cars: Waymo, Zoox, Tesla Full Self-Driving.

Sources: Gallup poll via WebProNews (2026); Futurism, Oct 9, 2026; Quanta Magazine, Apr 13 and Aug 3, 2026

WHAT NOW?

What Can You Do?

Be curious. Be studious. Be attentive.

Nobody knows exactly what to do. But by being here today, you've already started.

- **Keep informed:** follow the news; this field changes month to month.
- **Experiment:** try it on real tasks from your own life.
- **Learn its limits:** understand what it does well, and where it fails.
- **Speak up:** let your political leaders know how you feel.
- **Think it through:** what it means for you, your family and your community.

TWENTY WAYS I USE AI · 1 OF 3

Learning the Tool, Taming My Tech

#	Use case	Takeaway
1	**Explain each button**	A screenshot plus “explain this” = a guided tour.
2	**Which Claude is which?**	Ask the tool about its own settings and quirks.
3	**Getting my chat history out**	Your history is yours; it coached every step.
4	**A missing Mac window**	One running chat per device becomes your manual.
5	**Phone rang despite Sleep Focus**	Describe the problem; skip the setting's name.
19	**A teaching assistant “skill”**	Teach a weekly task once, then reuse it.
20	**Plugging into my other apps**	Connectors take it from answering to doing.

Source: my own Claude conversations, November 2025 to October 2026; full list in “Twenty Ways I Use Claude”

TWENTY WAYS I USE AI · 2 OF 3

Everyday Life and Decisions

#	Use case	Takeaway
8	**Should I open this attachment?**	A 30-second second opinion can stop a scam.
9	**Handouts or a projector?**	It recalls earlier research and does the math.
10	**Emergency medical information**	Prepare for the doctor; don't replace one.
11	**A 2 a.m. sleep coach**	A patient sounding board at any hour.
13	**Holland Tunnel at midnight**	Current conditions turned into a reusable plan.
17	**Is AppleCare worth $300?**	It lays out the numbers; the decision is yours.
18	**Kitchen scale, turkey thawing**	Everyday questions are some of the best uses.

Source: my own Claude conversations, November 2025 to October 2026; full list in “Twenty Ways I Use Claude”

TWENTY WAYS I USE AI · 3 OF 3

Making Things, and Checking Them

#	Use case	Takeaway
6	**Gray text to Hillel cue cards**	A small how-to can grow into a finished piece.
7	**Passport pages to travel diary**	Messy photos become organized, dated records.
12	**Health spreadsheet to an app**	Plain English can become software you own.
14	**A word coach before Send**	A partner for tone and word choice.
15	**Morning curiosity rabbit holes**	Half-remembered trivia, settled with sources.
16	**Trust, but verify**	Every AI errs: push back and cross-check.

Fun fact: this list was compiled by asking AI to search its own past chats with me.

Source: my own Claude conversations, November 2025 to October 2026; full list in “Twenty Ways I Use Claude”

May You Live in Interesting Times

THE “CHINESE CURSE”

“May you live in interesting times.”

No Chinese original has ever been found. The earliest known use is British statesman Austen Chamberlain in 1936, citing an unnamed diplomat.

ROBERT F. KENNEDY · CAPE TOWN, 1966

“Like it or not, we live in interesting times.”

Kennedy then turned the curse around: times of danger and uncertainty are also times of unusual creative possibility.

Where AI leaves us: squarely in interesting times, with great promise and real peril at once.

Sources: Quote Investigator, “May You Live in Interesting Times”; JFK Library, Day of Affirmation Address (1966)